

Turbulence: It is a fluid regime characterized by chaotic, stochastic property changes through rapid changes / variation of pressure and velocity in Space and time.

Vertical Transport / Buoyancy.

Net force exerted on an object, representing the difference between the PGF & gravity is called the buoyancy.

$$[\gamma_b = \gamma_p - g.]$$

$$F_b = \rho' v g - \rho v g$$

buoyant
upward
force

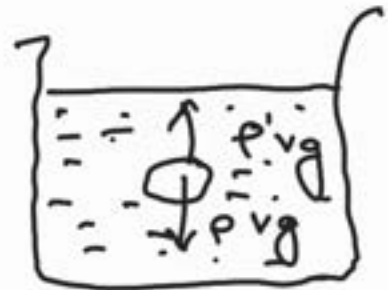
$$= v g (\rho' - \rho)$$

$$\frac{F_b}{m} = \frac{v g}{m} (\rho' - \rho)$$

$$\frac{F_b}{\rho v} = \frac{v g}{\rho v} (\rho' - \rho)$$

$$\Rightarrow \left[\gamma_b = \frac{(\rho' - \rho)}{\rho} \times g \right]$$

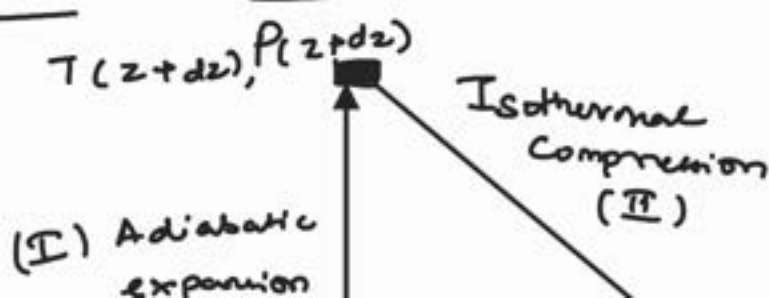
↳ Buoyancy / Buoyant acceleration

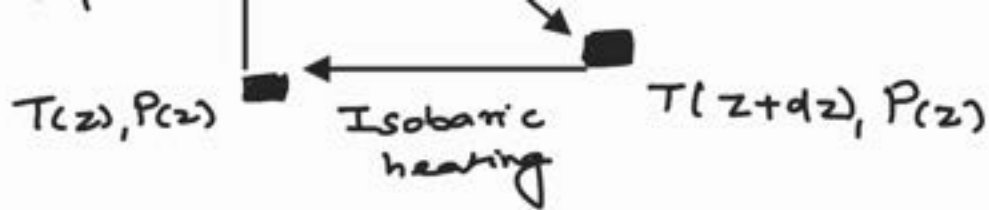


$$[\therefore \gamma_b = \rho' v g - \rho v g.]$$

$$[\rho = \frac{m}{V} \Rightarrow m = \rho V]$$

Adiabatic Lapce rate.





$$\times dH = dE + d(PV)$$

$$\times dH = dQ + dW + PdV + VdP$$

(I) Adiabatic Expansion, $dQ = 0$

$$dH_I = dE + d(PV)$$

$$= dQ - PdV + PdV + VdP \quad [\because dQ = 0]$$

$$dH_I = VdP \quad \text{--- (1)}$$

(II) Isothermal Compression, $dT = 0$

$$dH_{II} = dE + d(PV)$$

$$dH_{II} = 0 \quad \text{--- (2)}$$

$$\left[\begin{array}{l} \because dT = 0 \\ dE = 0 \\ PV = nRT \\ PdV = nRdT \\ PdV = 0 \end{array} \right]$$

(III) Isobaric heating, $dP = 0$

$$dH_{III} = dE + d(PV)$$

$$= dQ - PdV + VdP + PdV$$

$$dH_{III} = dQ$$

$$dH_{III} = -mC_p dT \quad \text{--- (III)}$$

$$\left[\begin{array}{l} \because m = \text{mass of the air parcel} \\ C_p = 1000 \text{ J K}^{-1} \text{ kg}^{-1} \\ \text{Specific heat capacity at constant pressure.} \end{array} \right]$$

For a cyclic process,

$$dH_I + dH_{II} + dH_{III} = 0$$

$$VdP + 0 - mC_p dT = 0$$

$$\Rightarrow VdP = mC_p dT$$

$$\rho = \frac{m}{V}$$

$$\Rightarrow VdP = \rho V C_p dT$$

$$\Rightarrow m = \rho \times V. \quad \left[\text{From Barometric law.} \right]$$

$$\Rightarrow \frac{dT}{dz} = \gamma$$

$$\left[\because \frac{dp}{dz} = -\rho g \right]$$

$$\Rightarrow \frac{dp}{dz} \times \frac{dz}{dT} = \rho \gamma$$

$$\Rightarrow -\rho g \times \frac{dz}{dT} = \rho \gamma$$

$$\Rightarrow \frac{dz}{dT} = -\frac{\rho \gamma}{\rho g}$$

or

$$-\frac{dT}{dz} = \frac{g}{\gamma} = \frac{9.8 \text{ m/s}^2}{1000 \text{ J K}^{-1} \text{ kg}^{-1}}$$

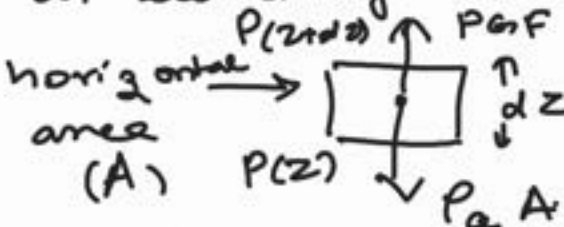
$$\gamma = \frac{9.8}{1000} \text{ K/m}$$

$$\gamma = \frac{9.8 \times 1000}{1000} \text{ K/km}$$

$$\boxed{\gamma = 9.8 \text{ K/km}}$$

BAROMETRIC LAW

Consider an elementary slab of atmosphere (thickness dz) horizontal area (A) at altitude (z) .



$$P.G.F = (P(z) - P(z+dz))A$$

For a slab to be in equilibrium, its weight must be in equilibrium, its weight must balance the pressure - gradient force.

$$\rho_a g A dz = (P(z) - P(z+dz))A$$

$$\Rightarrow \rho_a g = \frac{P(z) - P(z+dz)}{dz}$$

$$\Rightarrow \frac{P(z+dz) - P(z)}{dz} = -\rho_a g$$

$$\Rightarrow dp = -\rho_a g dz$$

Now, from Ideal gas law;

$$P_a = \frac{P M_a}{RT}$$

$$P_a = \frac{M_a}{V}$$

$$\left[P_a = \frac{P M_a}{RT} \right]$$

$$\Rightarrow \frac{dP}{dz} = - \frac{P M_a}{RT} \times g$$

$$\Rightarrow \int_{P(0)}^{P(z)} \frac{dP}{P} = - \frac{M_a g}{RT} \int_0^z dz$$

$$\Rightarrow \ln P \Big|_{P_0}^{P_z} = - \frac{M_a g}{RT} z$$

$$\Rightarrow \ln P(z) - \ln P(0) = - \frac{M_a g}{RT} z$$

$$\Rightarrow \ln \left\{ \frac{P(z)}{P(0)} \right\} = - \frac{M_a g}{RT} z$$

$$\Rightarrow P(z) = P(0) e^{-\frac{M_a g}{RT} z}$$

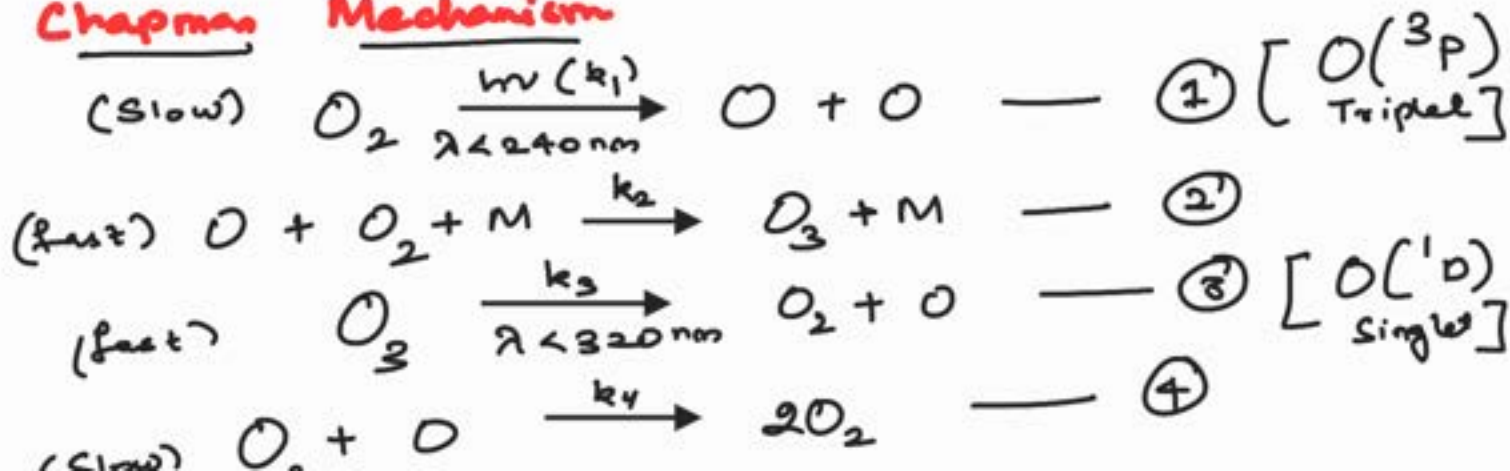
$$\Rightarrow P(z) = P(0) e^{-z/H} \quad \left[\because H = \frac{RT}{M_a g} \right]$$

For a mean atmospheric temperature $T = 250K$, the Scale height is $H = 7.4km$.

STRATOSPHERIC OZONE

The Stratospheric Ozone layer, centered at about 20km above the surface of the Earth, protects the life on Earth by absorbing UV radiation from the Sun.

Chapman Mechanism



R_2, R_3 are fast reactions
 R_1, R_4 are slow reactions

Because of rapid cycling between O and O_3 it is convenient to refer to the sum of the two as a chemical family, odd oxygen ($O_x \equiv O_3 + O$)

Step 1 Lifetime of O atom τ_O

$$\tau_O = \frac{[O]}{\text{Loss rate of } O} = \frac{[O]}{R_2 + R_4}$$

Since $R_2 \gg R_4$.

$$= \frac{[O]}{R_2} = \frac{[O]}{k_2 [O] [O_2] M}$$

$$\left[\begin{array}{l} \therefore [O_2] = C_{O_2} n_a \\ [M] = n_a \end{array} \right] = \frac{1}{k_2 [O_2] M}$$

$$\tau_O = \frac{1}{k_2 C_{O_2} n_a^2} < 1 \text{ sec}$$

Step 2

Assuming chemical steady state.

Rate of formation = Rate of loss

$$R_1 = R_3$$

$$k_2 [O] [O_2] [M] = k_3 [O_3]$$

$$\Rightarrow \frac{[O]}{[O_3]} = \frac{k_3}{k_2 [O_2] [M]} = \frac{k_3}{k_2 C_{O_2} n_a^2}$$

$$\Rightarrow \frac{[O]}{[O_3]} \ll 1 \text{ throughout the stratosphere}$$

This suggests that in stratosphere O_3 is the main component of O_x family

Step 3

Lifetime of O_x . $\{[O_x] \approx [O_3]\}$

$$\tau_{O_x} = \frac{[O_x]}{2 k_4 [O_3] [O]} = \frac{1}{2 k_4 [O]}$$

Step 4

$$[O_x] = [O_3] + [O]$$

$$\frac{d[O_x]}{dt} = \frac{d[O_3]}{dt} + \frac{d[O]}{dt}$$

$$= (R_2 - R_3 - R_4) + (2R_1 - R_2 + R_3 - R_4)$$

$$\frac{d[O_x]}{dt} = 2R_1 - 2R_4$$

Assuming steady state approximation,

$$\frac{d[O_x]}{dt} = 0$$

$$\Rightarrow 2R_1 = 2R_4$$

$$\Rightarrow k_1 [O_2] = k_4 [O_3][O]$$

$$[O_3] = \frac{k_1 [O_2]}{k_4 [O]} \left[\because \frac{[O]}{[O_3]} = \frac{k_3}{k_2 [O_2] n_a^2} \right]$$

$$[O_3] = \frac{k_1 [O_2]}{k_4 \left(\frac{k_3 [O_3]}{k_2 [O_2] n_a^2} \right)}$$

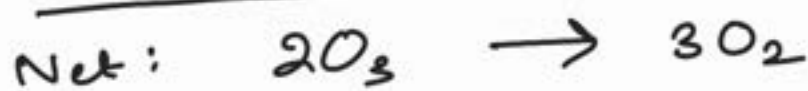
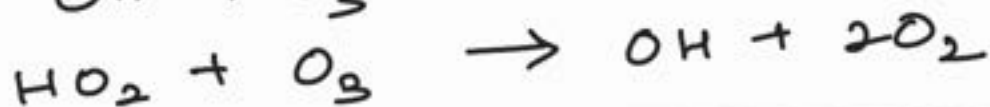
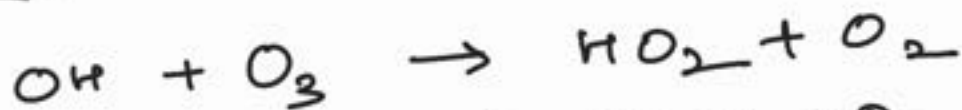
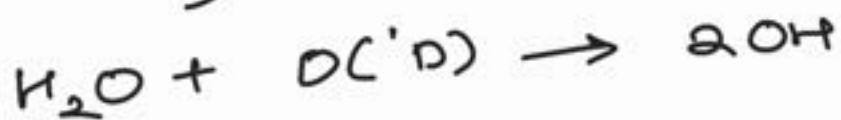
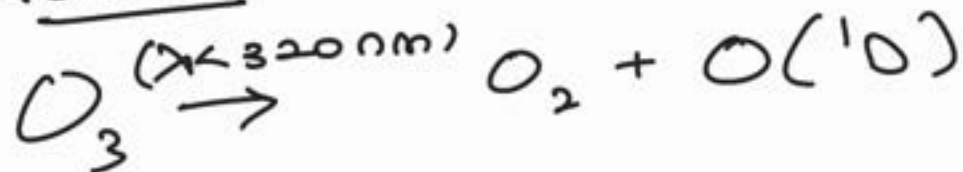
$$\Rightarrow [O_3]^2 = \frac{k_1 k_2}{k_3 k_4} \cdot [O_2]^2 \cdot n_a^3$$

$$\Rightarrow \boxed{[O_3] = \left(\frac{k_1 k_2}{k_3 k_4} \right)^{1/2} [O_2] n_a^{3/2}}$$

Although the Chapman mechanism is successful in reproducing the general shape of the O_3 layer, it overestimates the O_3 concentrations by a factor of 2 or more.

CATALYTIC LOSS CYCLES.

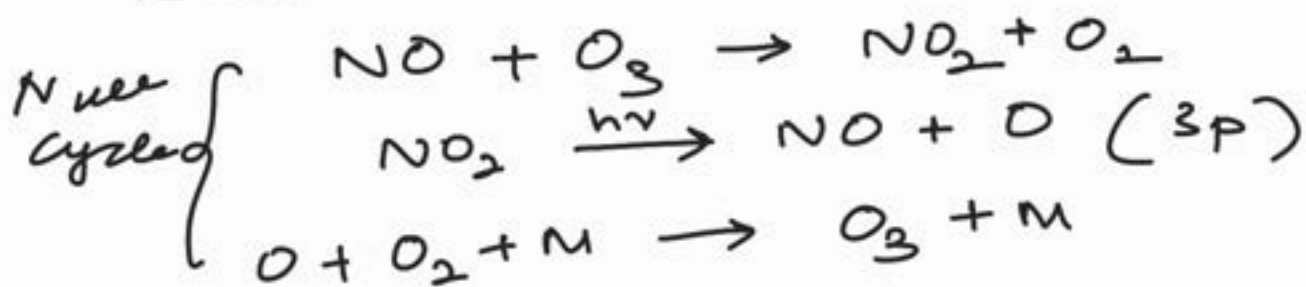
1. HO_x radicals



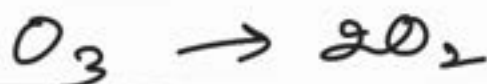
Termination



2. Nitrogen oxide radicals (NO_x)



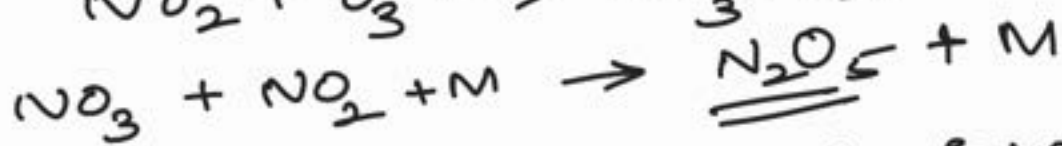
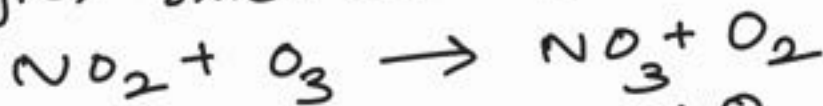
Further a fraction of NO₂ produced



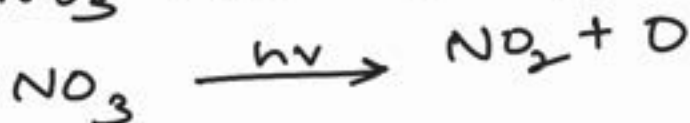
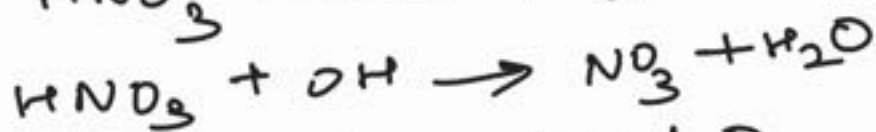
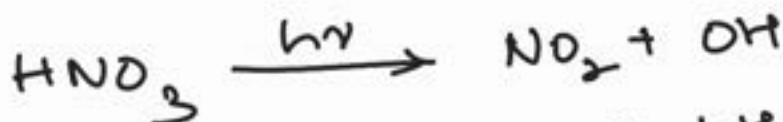
In Daytime, NO_x is oxidized to HNO₃ by strong radical oxidant OH.



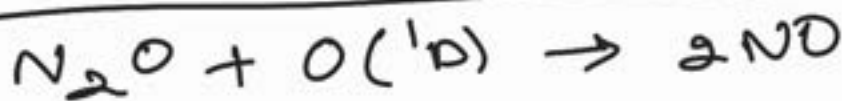
At Night, since OH is absent.



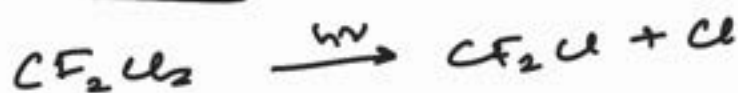
formation of N_2O_5 is favourable only during night as during daytime the NO_3 rapidly photolyzes to NO_2

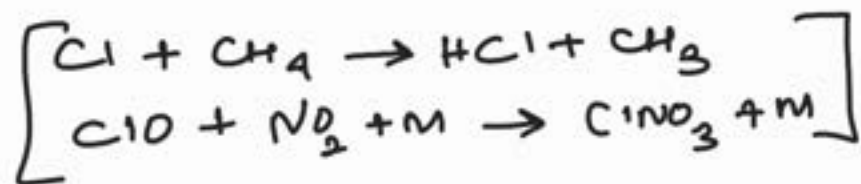


N_2O , a low yield product of nitrification and denitrification processes in the biosphere provides as a source of NO_x in the Stratosphere

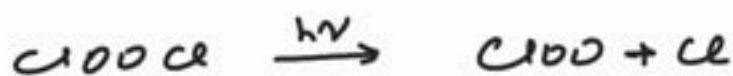


3. Chlorine radicals. (ClO_x)

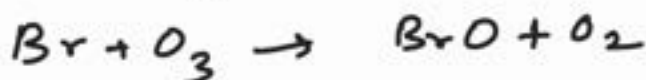
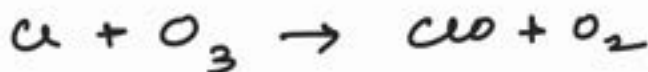




Mechanism of O₃ loss.



Br radicals produced in the stratosphere during antarctic spring by photolysis and oxidation of anthropogenic gases such as CH_3Br

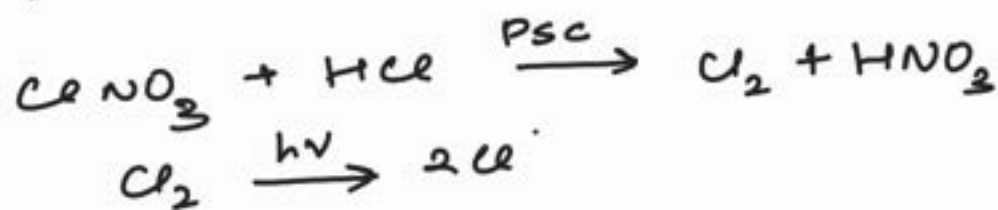


According to the current models, the $\text{ClO} + \text{ClO}$ mechanism accounts for 70% of total O_3 loss in the antarctic region, the remaining 30% is reduced by the $\text{BrO} + \text{ClO}$ mechanism

Ozone hole only in antarctic region???

Temperatures in the wintertime antarctic stratosphere

are sufficiently cold to cause formation of persistent ice-like clouds called Polar Stratospheric clouds (PSC) in the lower part of the stratosphere. The PSCs provide surfaces for conversion of the ClO_x reservoirs HCl and ClONO_2 to Cl_2 , which then rapidly photolyzes to produce ClO_x .



Problem:

10.2 The Chapman mechanism & Steady state.

Here we compare some features of Chapman mechanism at 20 km and 45 km altitude. Adopt temperature of 200 K at 20 km altitude and 270 K at 45 km.

$$n_a(20\text{ km}) : 1.8 \times 10^{18} \text{ molecules cm}^{-3}$$

$$n_a(45\text{ km}) : 4.1 \times 10^{16} \text{ molecules cm}^{-3}$$

Given $k_2 = 1 \times 10^{-33} \text{ cm}^6 \text{ molecule}^{-2} \text{ s}^{-1}$

$$k_3 = 1 \times 10^{-2} \text{ s}^{-1}$$

$$k_4 = 8.0 \times 10^{-12} \exp(-2060/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$$

Q1 To calculate the Lifetime of O atom at 20 km and 45 km altitude

$$\tau_{\text{O}} = \frac{[\text{O}]}{k_2 [\text{O}] [\text{O}_2] [\text{M}]} = \frac{1}{k_2 [\text{O}_2] [\text{M}]}$$

$$= \frac{1}{k_2 [\text{O}_2] n_a^2}$$

$$\tau_{\text{O}}(20\text{ km}) = \frac{1}{10^{-33} \times 0.21 \times (1.8 \times 10^{18})^2}$$

$$= \frac{1}{0.6804 \times 10^{-3}}$$

$$= 1.469 \times 10^3$$

$$= 1.5 \times 10^3 \text{ sec.}$$

$$J_0 = \frac{1}{10^{-23} \times 0.21 \times (4.1 \times 10^{16})^2} = 2.8 \text{ sec}$$

(45km)

The Oxygen atom can be assumed to be in chemical steady state throughout the Stratosphere because its chemical lifetime is so short relative to the time scales of transport

Q2. $\frac{[O]}{[O_3]} = ?$ at 20km and 95 km.

Assuming Steady State approximation,

$$R_1 = R_3$$

$$k_2 [O][O_2][M] = k_3 [O_3]$$

$$\Rightarrow \frac{[O]}{[O_3]} = \frac{k_3}{k_2 [O_2][M]}$$

$$\frac{[O]}{[O_3]} = \frac{k_3}{k_2 C_{O_2} n_a^2}$$

@ 20km; $\frac{[O]}{[O_3]} = \frac{1 \times 10^{-2}}{1 \times 10^{-23} \times 0.21 \times (1.8 \times 10^{18})^2}$

mol/mol $\leftarrow$

$$= 1.5 \times 10^{-5} \text{ mol/mol.}$$

@ 95km; $\frac{[O]}{[O_3]} = \frac{1 \times 10^{-2}}{1 \times 10^{-33} \times 0.21 \times (4.1 \times 10^{16})^2}$

$$= 0.028 \text{ mol/mol}$$

Yes, we can assume $[O_3] = [O_x]$ throughout the Stratosphere

Q3 To show that the mass balance equation for odd oxygen ($O_x = O_3 + O$), can be written as

$$\frac{d[O_x]}{dt} = P - k[O_x]^2$$

where $P = 2k_1[O_2] \rightarrow O_x$ production rate

$$k = 2k_3k_4/k_2C_{O_2}n_a^2$$

Mass balance equation for O_x .

$$\begin{aligned}\frac{d[O_x]}{dt} &= \frac{d[O_3]}{dt} + \frac{d[O]}{dt} \\ &= (R_2 - R_3 - R_4) + (2R_1 - R_4 - R_2 + R_3) \\ \frac{d[O_x]}{dt} &= 2R_1 - 2R_4\end{aligned}$$

$$\frac{d[O_x]}{dt} = 2k_1[O_2] - 2k_4[O_3][O]$$

$$\frac{d[O_x]}{dt} = P - \frac{2k_4[O_3][O_3]k_3}{k_2C_{O_2}n_a^2} \left[\because \frac{[O]}{[O_3]} = \frac{k_3}{k_2C_{O_2}n_a^2} \right]$$

$$\frac{d[O_x]}{dt} = P - \frac{2k_3k_4}{k_2C_{O_2}n_a^2} \cdot [O_x]^2$$

$$\Rightarrow \boxed{\frac{d[O_x]}{dt} = P - k[O_x]^2}$$

Q4

$[O_x]$

1

$$J_{O_2} = \frac{k_2 C_{O_2} n_a^2}{2 k_4 [O_3] [O]} = \frac{k_2 C_{O_2} n_a^2}{2 k_4 [O]}$$

$$J_{O_2} = \frac{k_2 C_{O_2} n_a^2}{2 k_3 k_4 [O_3]}$$

$$[O] = \frac{k_3 [O_3]}{k_2 C_{O_2} n_a^2}$$

$$[O_3] = \left(\frac{k_1 k_2}{k_3 k_4} \right)^{1/2} C_{O_2} n_a^{3/2}$$

$$\Rightarrow \frac{[O]}{[O_3]} = 1.5 \times 10^{-5} \text{ mol/mol}$$

$$\Rightarrow \frac{[O_3]}{[O]} = \frac{1}{1.5 \times 10^{-5}} = 0.67 \times 10^5 \frac{\text{mole}}{\text{mole}}$$

@ 20 km,

$$n_x = C_x n_a$$

$$= 0.67 \times 10^5 \times 1.8 \times 10^{20} \text{ molecules/cm}^3$$

$$n_x = 1.206 \times 10^{25} \text{ molecules/cm}^3$$

$$J_{O_2} = \frac{[O_2]}{k [O_2]^2} = \frac{1}{k [O_2]}$$

10.3 Detailed Chapman Mechanism.

$$\textcircled{1} J_{O(^1O)} = \frac{[O(^1O)]}{L [O(^1O)]} = \frac{[O(^1O)]}{k_5 [O(^1O)] [N_2] + k_6 [O(^1O)] [O]}$$

$$J_{O(^1D)} = \frac{[O(^1D)]}{[O(^1D)] \left\{ \frac{k_5 [N_2]}{k_6 [O_2]} \right\} + 1}$$

$$= \frac{1}{k_5 [N_2] + k_6 [O_2]}$$

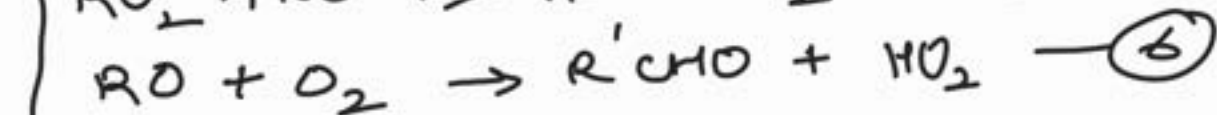
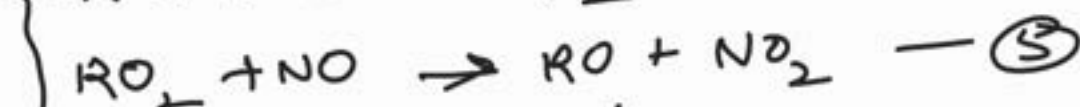
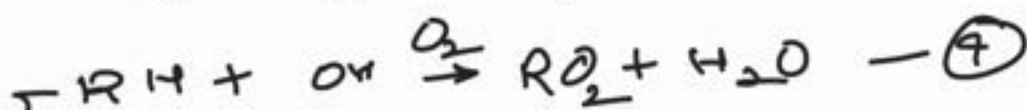
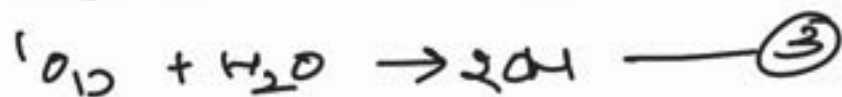
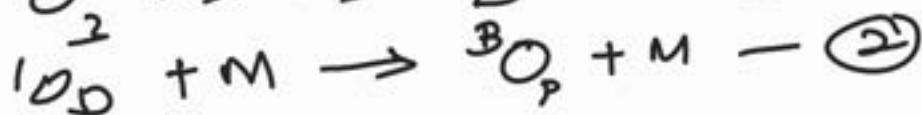
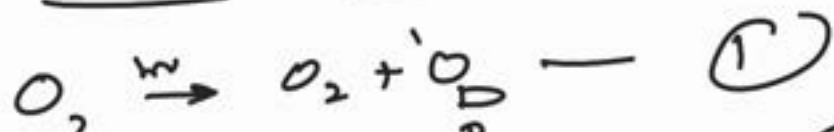
Lifetime of O_x should be calculated from the equation when O_3 & O reacts with each other.

$$L_{O_x} = P_{O_x} = 2k_1 [O_2]$$

Loss from the Chapman mechanism by reaction (7) accounts for a fraction.

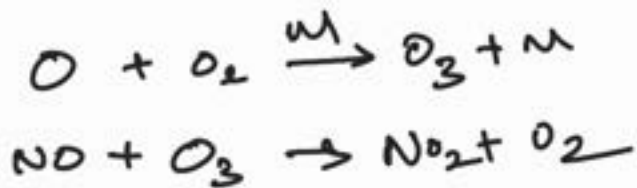
$$\frac{2k_7 [O(^3P)] [O_3]}{2k_1 [O_2]} = 24\% \text{ of total } O_x \text{ loss}$$

Tropospheric Ozone

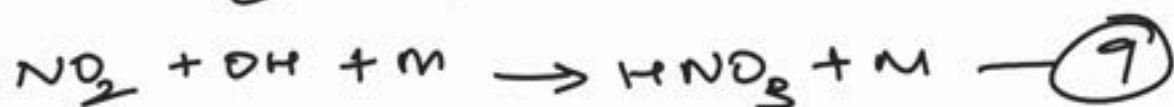


Propagation

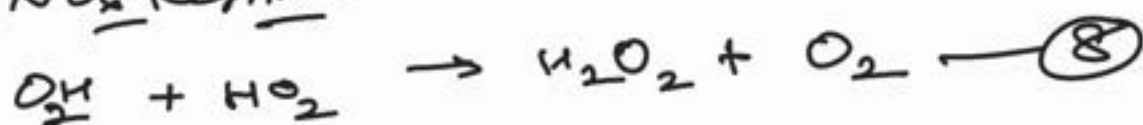
NOx cycle



Chain Termination, (1) High NOx Regime



(2) Low NOx Regime



Case Δ Low NOx

$$P_{O_3} = R_5 + R_7$$

$$= k_5 [AO_2] [NO] + k_7 [HO_2] [NO]$$

$$P_{O_3} = 2R_7 \quad [\because R_5 = R_7 = R_4 = R_6 \text{ all are chain propagation steps and occur at same rate}]$$

$$P_{O_3} = 2k_7 n_{NO} n_{HO_2} \quad \text{--- (I)}$$

$$P_{HO_x} = L_{HO_x} = 2k_8 n_{HO_2}^2$$

$$\Rightarrow n_{HO_2} = \left(\frac{P_{HO_x}}{2k_8} \right)^{1/2} \quad \text{--- (II)}$$

By substituting eqⁿ II in eqⁿ I, we get

$$P_{O_3} = 2k_7 \cdot n_{NO} \cdot \left(\frac{P_{HO_x}}{2k_8} \right)^{1/2}$$

$$P_{O_3} \propto n_{NO}$$

II. High NO_x

$$P_{O_3} = 2k_2 n_{NO} n_{NO_2}$$

$$P_{NO_x} = L_{NO_x} = k_9 n_{NO_2} n_{OH} [M] \xrightarrow{NO_2 + OH + M} HNO_2 + M$$

→ (forming SSA)

$$\boxed{R_4 = R_2}$$

$$k_4 [RH] [OH] = k_2 [HNO_2] [NO]$$

$$\Rightarrow [HNO_2] = \frac{k_4 [RH] [OH]}{k_2 [NO]}$$

$$P_{O_3} = 2k_2 [NO] \left\{ \frac{k_4 [RH] [OH]}{k_2 [NO]} \right\}$$

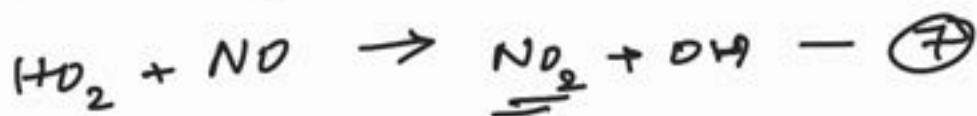
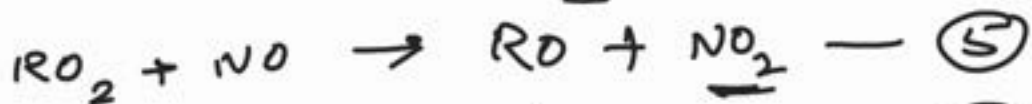
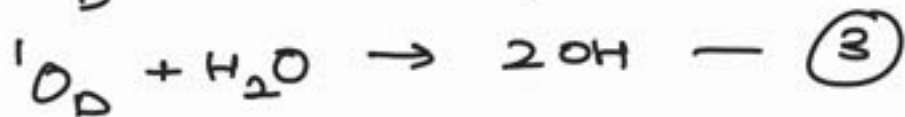
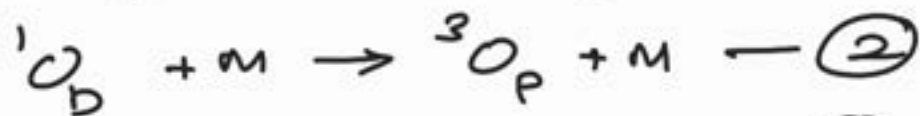
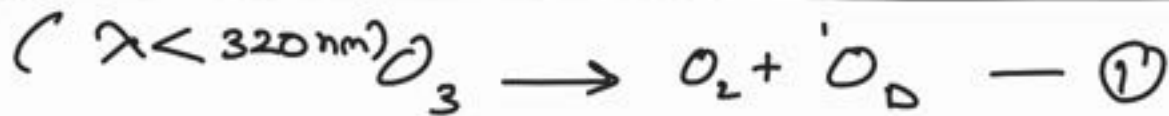
$$P_{O_3} = 2k_4 [RH] [OH]$$

$$[OH] = \frac{P_{NO_x}}{k_9 [NO_2] [M]}$$

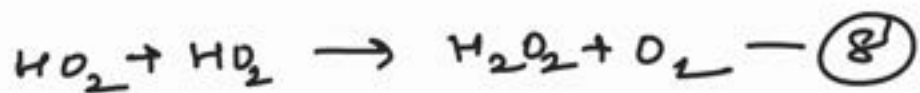
$$P_{O_3} = \frac{2k_4 [RH] \cdot P_{NO_x}}{k_9 [NO_2] [M]}$$

$$P_{O_3} \propto [RH]$$

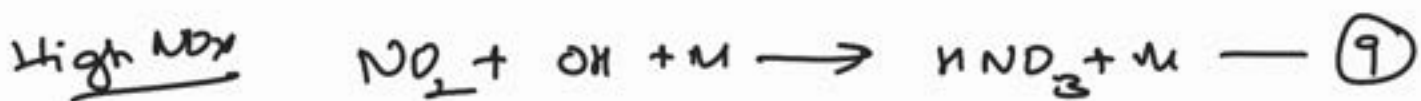
$$PO_3 \propto \frac{1}{NOx}$$



Low NOx

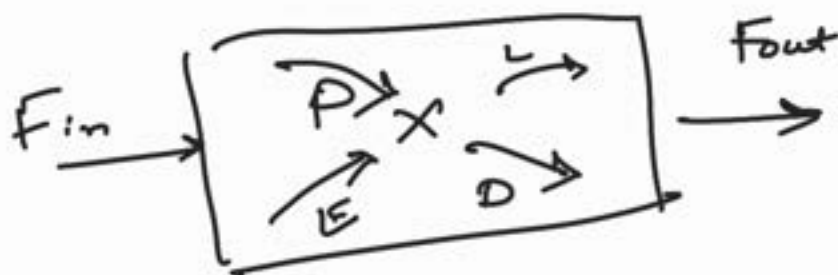


High NOx



Simple Box Model

Life time, Residence time, Inflows, Removal



$$\frac{dM_x}{dt} = \sum \text{Source} - \sum \text{Sink}$$

$$\frac{dM_{xo}}{dt} = E + P - L - D - \text{Fouling} + F_{in}$$

Source term
Constant

$$\frac{dM_{xo}}{dt} = S - km$$

$$k = k_L + k_D + k_{out}$$

$$J = \frac{m}{L + D + \text{Fouling}}$$

So k
our loss rate
is km

$$= \frac{m}{k_L m + k_D m + k_{out} m}$$

$$J = \frac{1}{k_L + k_D + k_{out}} = \frac{1}{k}$$

$$\frac{dm}{dt} = S - km$$

$$\int \frac{dm}{S - km} = \int dt$$

$$z = S - km$$

$$dz = -k dm$$

$$dm = \frac{dz}{-k}$$

$$\Rightarrow \int \frac{dz}{-k z} = t \Big|_0^t$$

$$\Rightarrow -\frac{1}{k} [\ln(S - km(t)) - \ln(S - km(0))] = t$$

$$\Rightarrow \ln \frac{S - km(t)}{S - km(0)} = -kt$$

$$\Rightarrow \frac{S - km(t)}{S - km(0)} = \frac{-kt}{e}$$

$$-kt$$

$$\Rightarrow S - km(t) = (S - km(0)) e^{-kt}$$

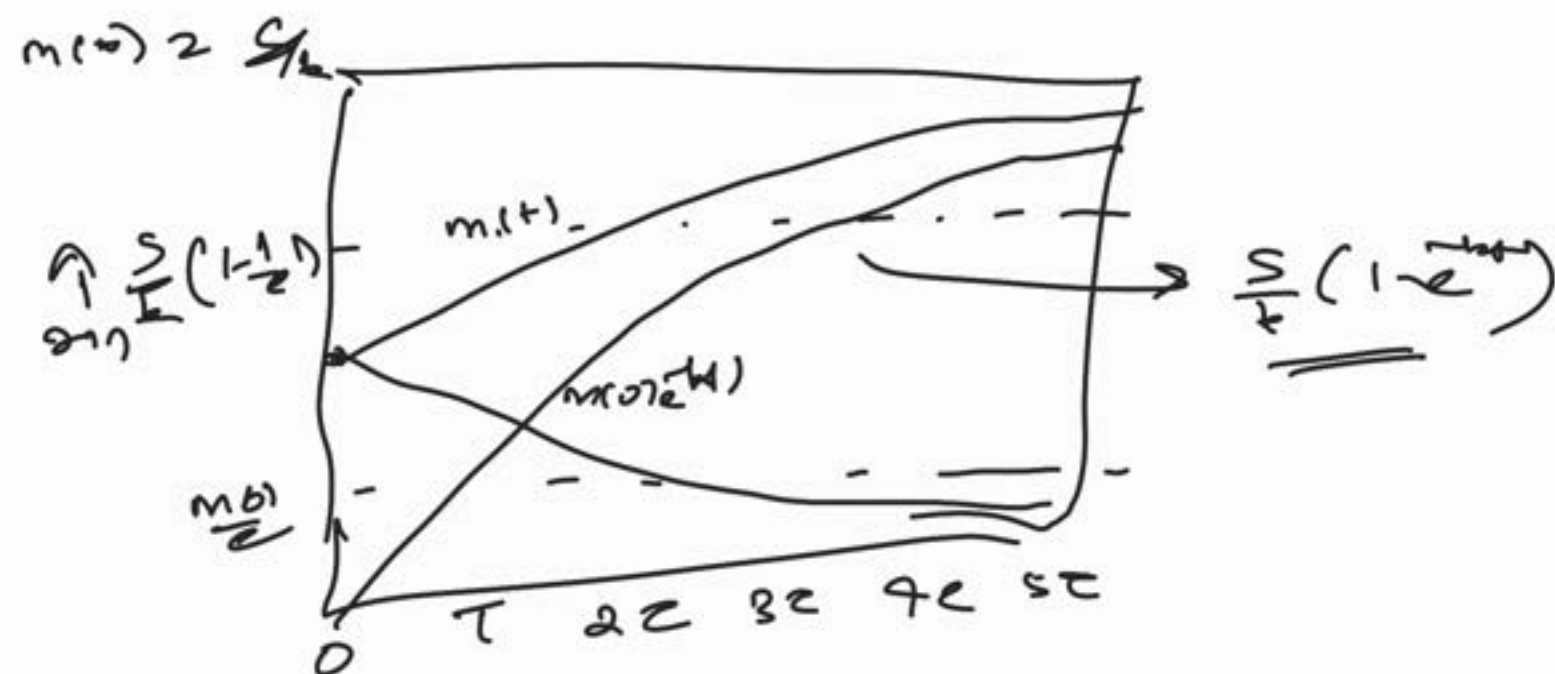
[Dividing by $-k$]

$$\Rightarrow \frac{S}{-k} - \frac{km(t)}{-k} = \frac{S - km(0)}{-k} e^{-kt}$$

$$m(t) = m(0) e^{-kt} + \frac{S}{k} (1 - e^{-kt})$$

$$t = \infty \quad m(\infty) = S/k \quad (y, 1/k)$$

$$t = T \quad m(T) = \frac{m(0)}{e} + \frac{S}{k} (1 - \frac{1}{e})$$



Pubb Model

$$\frac{dSx}{dt} = E + P - L - D - k_d(x - x_b)$$

Column model,

$$\frac{d[x]}{dt} = \frac{E}{h} + P - L - D$$

Assume $P = 0$
First order L



S-km

$$\frac{d[x]}{dt} = \frac{E}{h} - k[x]$$

$$\frac{d[x]}{dx} \cdot \frac{dx}{dt} = \frac{E}{h} - k[x]$$

$$x \frac{d[x]}{dx} \cdot U = \frac{E}{h} - k[x]$$

$$\int_0^x \frac{d[x]}{\frac{E}{h} - k[x]} = \int_0^x \frac{dx}{U}$$

$$\Rightarrow -\frac{1}{k} \left(\frac{E}{h} - k[x] \right)_0^x = \frac{x}{U}$$

$$\Rightarrow \ln \left(\frac{E}{h} - k[x] \right) - \ln \frac{E}{h} = -\frac{kx}{U}$$

$$\Rightarrow \frac{\frac{E}{h} - k[x]}{\frac{E}{h}} = e^{-kx/U}$$

$$\Rightarrow E - k[x] = \frac{E}{h} \cdot e^{-kx/U}$$

$$\Rightarrow k[x] = \frac{E}{h} - \frac{E}{h} e^{-kx/v}$$

$$\Rightarrow k[x] = \frac{E}{h} (1 - e^{-kx/v})$$

$$\Rightarrow \sqrt{[x]} = \frac{E}{hk} [1 - e^{-kx/v}]$$

$$\textcircled{1} x < 0, \quad E: 0 \rightarrow \infty$$

$$\textcircled{2} 0 \leq x < L \quad \checkmark$$

$$\textcircled{3} x \geq L \quad [x] = [x]_L e^{-\frac{k(x-L)}{v}}$$

$$\frac{d[x]}{dt} = \frac{E}{h} - k[x]$$

$$\frac{d[x]}{dt} = -k[x]$$

$$\frac{d[x]}{dx} \cdot \frac{dx}{dt} = -k[x]$$

$$\frac{d[x]}{dx} \cdot v = -k[x]$$

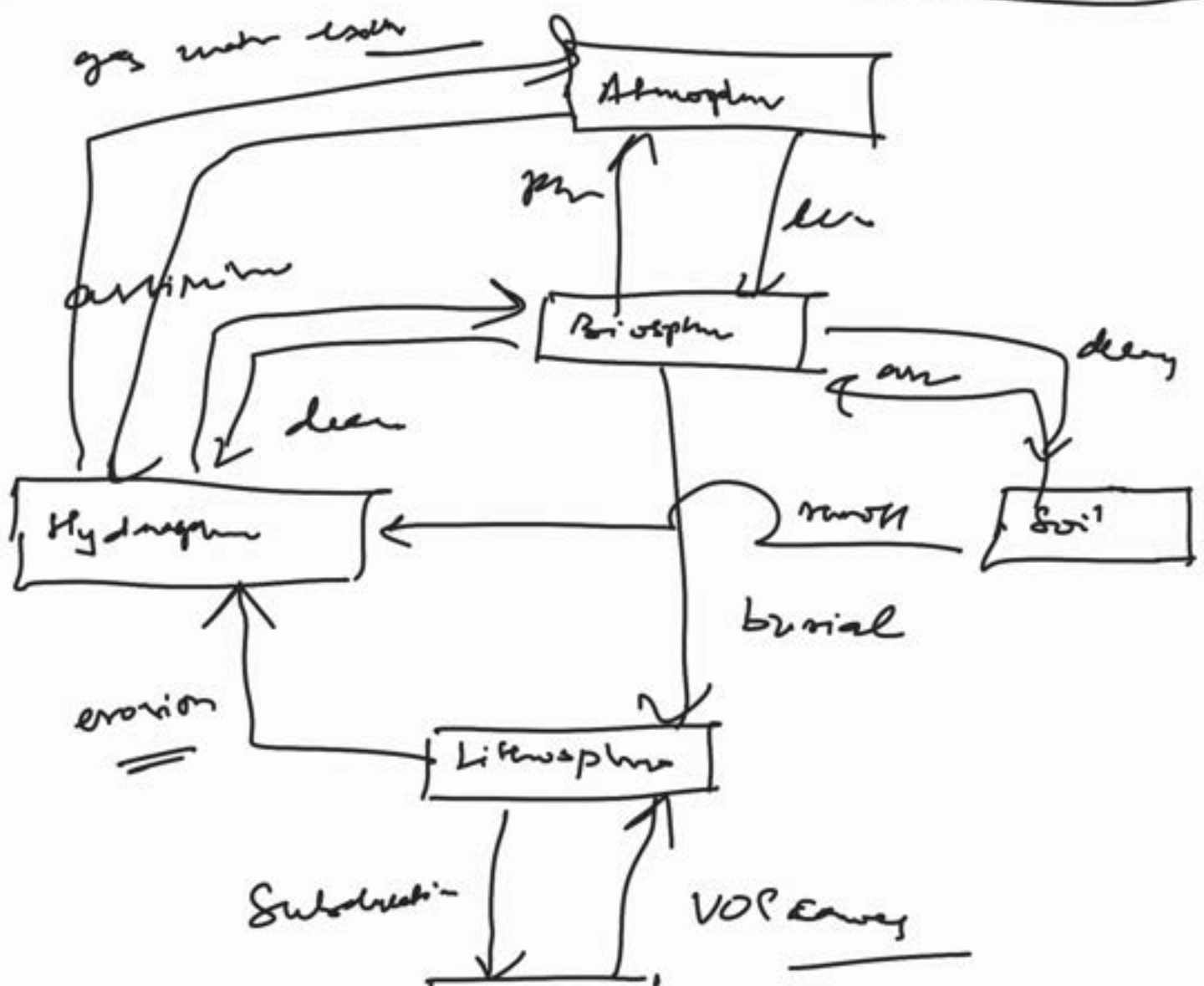
$$\frac{d[x]}{[x]} = \frac{-k}{v} \cdot dx$$

$$\ln [X] \Big|_{x_L}^x = - \frac{k}{v} \cdot \Big|_{L}^x$$

$$\ln [X] - \ln [X]_L = - \frac{k}{v} (x - L)$$

$$\ln \frac{[X]}{[X]_L} = - \frac{k(x-L)}{v}$$

$$[X] = [X]_L e^{-\frac{k(x-L)}{v}}$$



Deep
Earth
